⇒ Multi-Agent Architecture.

↓

Perceive – Reason – Act.

⇒ Multi-Agent architecture is an architecture where multiple specialized agents work together to solve a particular task.

Problems with Single Agent architecture

→ Too many responsibilities.

→ Weak Observability.

→ Difficult to scale.

Core Components of Multi-Agent architecture

User Request

↓

Coordinator | Manager | Supervisor

↓

Specialized Agents

↓

Shared Memory

↓

Tools | APIs | Databases

↓

Verifier | Critic | Human Review

↓

Final Response

→ Agent

→ Coordinator | Supervisor Agent.

Shared Memory | State

↓

Allows the agents to access common information.

→ Without shared memory, agents may get stuck or may repeat some task.

Tools.

↳ External call to take some action.

→ Ex: DB | Email | Book flight | hotel | process refund ....

Multi-Agent architecture patterns.

1. Supervisor | Manager | Coordinator Pattern.

↳ One central agent which manages all other agents.

2. Sequential Pipeline Pattern

↳ Agents work in a particular order, one after another.

Input → Research Agent → Writer Agent → Editor Agent → Reviewer Agent → Uploader Agent

5. Hierarchical Pattern.

↳ Agents are organized into different levels.

Top-Level Manager
↓
Domain Manager
↓
Specialized Agents.

Communication Protocols.

↳ How Agents will exchange information.

1. Direct Message Passing

↳ One agent sends a message directly to other agent.

Coordinator Agent : Refund agent.

↳ Sends a msg directly to Refund agent

Please check the refund status for orderId : 1234.

↓

Refund agent

Response.

```
{
  "agent" : "RefundAgent",
  "status" : "in_progress",
  "reason" : "_____"
  "___" : "_____"
}
```

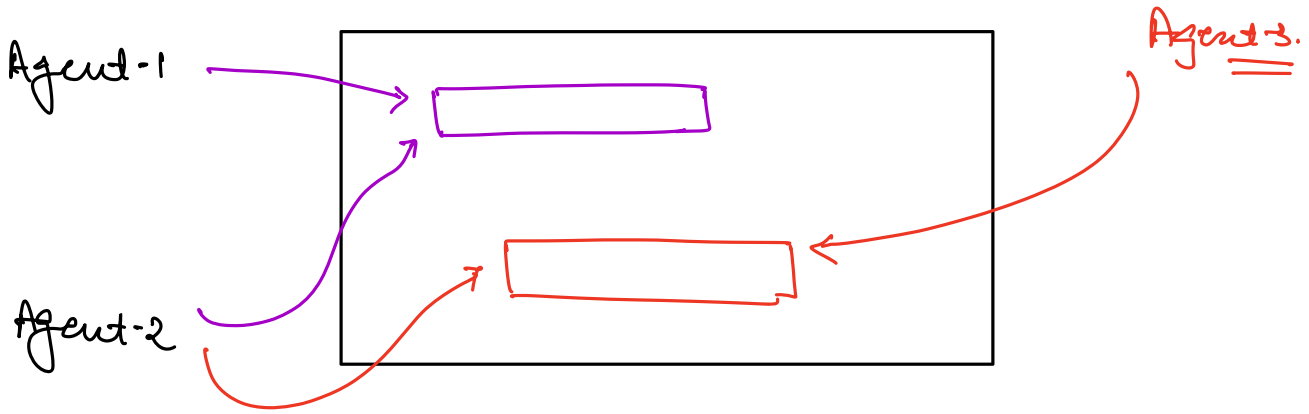
Handoff protocol.

↳ Control moves from one agent to another.

Research Agent → Writer Agent.

Shared Memory Protocol.

→ Agents communicate indirectly through shared memory.



Task Delegation & Coordination.

Shared Memory across Agents.

Short Term Memory
↳ During the Conversation.

Long-Term Memory

Consensus Mechanisms.

↓

Means multiple agents agree on a decision.

→ Majority Voting

→ Weighted Voting

Judge | Critic Agent.

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